



United States Environmental Protection Agency
Office of Enforcement and Compliance Assurance
 1200 Pennsylvania Ave NW, Mail Code 2227A
 Washington, DC 20460

DMR-QA Study 38

(This data is collected under the authority of Section 308 of the Clean Water Act.)

NPDES Permittee Data Report Form

Attention: Follow the instructions on the previous page to complete this form and submit data for evaluation.

Due August 31, 2018

State

NPDES Permit Number

Permit Extension

MI

0020257

Permittee Name

Cadillac WWTP

Current Permittee Mailing Address

200 N Lake St

City

Cadillac

State

MI

Zip Code

49601

Phone Number

+1 (231) 775-2368

Fax Number

+1 (231) 775-2876

E-mail

lab@cadillac-mi.net

Optional: If WP Study was used, list PT Provider name(s):

Optional: WP Study Number(s)

For DMR-QA Study 38, conducted in 2018, the Permittee ensured that their laboratory(ies) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data by July 6, 2018

Yes ☒ No ☐

Received a Graded Report by August 3, 2018

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative

(as per 40 CFR Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) permit. Neither I nor any of my subordinates compared our results with results from independent analyses conducted by us or any other laboratory before we reported our results to the U.S.EPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of Certifying Official

Cindy A. Tomaszewski

Signature

Title

Lab Supervisor

Date

8/31/2018

Address, phone number and e-mail of certifying official are required if different from above.

Address

Phone Number

City

State

Zip Code

E-mail



United States Environmental Protection Agency

Office of Enforcement and Compliance Assurance

1200 Pennsylvania Ave NW, Mail Code 2227A

Washington, DC 20460

DMR-QA Study 38

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Permittee Name

State

NPDES Permit No.

Permit Extension

Cadillac WWTP

MI

0020257

Identification of all CHEM, MICRO and WET laboratories who performed analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			Chem	Micro	WET		
Pace Analytical Services	5560 Corporate Exchange Ct SE Grand Rapids MI 49512	MI00005	☒	☐	☐	C	☐
City of Cadillac Contract Lab	1121 Plett Rd Cadillac MI 49601	MI00082	☒	☒	☐	G	☐
City of Cadillac Wastewater Lab	1121 Plett Rd Cadillac MI 49601	MI01099	☒	☐	☐	G	☐
Environmental Resources Management	3352 128th Ave Holland MI 49424	MI01183	☐	☐	☒	C	☐
			☐	☐	☐		☐
			☐	☐	☐		☐

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote on page 2 (Frequently Asked Questions) for the current list of states with lab accreditation programs

If you need additional space, please make a copy of this page for additional laboratories.

Chemistry/Microbiology Analyte Checklist

DMR-QA Study 38

Analyte Test	Test Required	Method Number Used (optional)	Laboratory's Graded Result		Analyte determined by state-certified lab*
			Acceptable	Not Acceptable (Corrective Action Required)	
Microbiology					
<i>E. coli</i> , MF or MPN	☐		☐	☐	☐
Fecal Coliform, MF or MPN	☐		☐	☐	☐
Total Coliform, MF or MPN	☐		☐	☐	☐
Trace Metals					
Aluminum	☐		☐	☐	☐
Antimony	☐		☐	☐	☐
Arsenic	☐		☐	☐	☐
Barium	☐		☐	☐	☐
Beryllium	☐		☐	☐	☐
Cadmium	☐		☐	☐	☐
Chromium, total	☐		☐	☐	☐
Chromium, hexavalent	☐		☐	☐	☐
Cobalt	☐		☐	☐	☐
Copper	☐		☐	☐	☐
Iron	☐		☐	☐	☐
Lead	☐		☐	☐	☐
Manganese	☐		☐	☐	☐
Mercury	☒		☒	☐	☐
Mercury (Low Level)	☐		☐	☐	☐
Molybdenum	☐		☐	☐	☐
Nickel	☐		☐	☐	☐
Selenium	☐		☐	☐	☐
Silver	☐		☐	☐	☐
Thallium	☐		☐	☐	☐
Vanadium	☐		☐	☐	☐
Zinc	☐		☐	☐	☐
Demands					
5-day BOD	☐		☐	☐	☐
5-day Carbonaceous BOD	☐		☐	☐	☐
COD	☐		☐	☐	☐
TOC	☐		☐	☐	☐
Minerals					
Alkalinity, total (CaCO ₃)	☐		☐	☐	☐
Chloride	☐		☐	☐	☐
Fluoride	☐		☐	☐	☐
Hardness, total (CaCO ₃)	☐		☐	☐	☐
Specific conductance (25°C)	☐		☐	☐	☐
Sulfate	☐		☐	☐	☐
Total Dissolved Solids (180°C)	☐		☐	☐	☐
Nutrients					
Ammonia as N	☐		☐	☐	☐
Nitrate as N	☐		☐	☐	☐
Nitrite as N	☐		☐	☐	☐
Orthophosphate as P	☐		☐	☐	☐
Total Kjeldahl-Nitrogen as N	☐		☐	☐	☐
Total Phosphorus as P	☐		☐	☐	☐
Misc. Analytes					
Non-Filterable Residue (TSS)	☐		☐	☐	☐
Oil and Grease	☐		☐	☐	☐
pH	☐		☐	☐	☐
Total Cyanide	☐		☐	☐	☐
Total Phenolics (4-AAP)	☐		☐	☐	☐
Total Residual Chlorine	☐		☐	☐	☐
Total Residual Chlorine (Low Level)	☐		☐	☐	☐
Settleable Solids	☐		☐	☐	☐
Turbidity	☐		☐	☐	☐

Name Cindy A. TomaszewskiSignature C. A. TomaszewskiDate 08/31/2018

Chemistry/Microbiology Analyte Checklist

DMR-QA Study 38

Analyte Test	Test Required	Method Number Used (optional)	Laboratory's Graded Result		Analyte determined by state-certified lab*
			Acceptable	Not Acceptable (Corrective Action Required)	
Microbiology					
<i>E. coli</i> , MF or MPN	☐		☐	☐	☐
Fecal Coliform, MF or MPN	☐		☐	☐	☐
Total Coliform, MF or MPN	☐		☐	☐	☐
Trace Metals					
Aluminum	☐		☐	☐	☐
Antimony	☐		☐	☐	☐
Arsenic	☐		☐	☐	☐
Barium	☐		☐	☐	☐
Beryllium	☐		☐	☐	☐
Cadmium	☐		☐	☐	☐
Chromium, total	☐		☐	☐	☐
Chromium, hexavalent	☐		☐	☐	☐
Cobalt	☐		☐	☐	☐
Copper	☐		☐	☐	☐
Iron	☐		☐	☐	☐
Lead	☐		☐	☐	☐
Manganese	☐		☐	☐	☐
Mercury	☐		☐	☐	☐
Mercury (Low Level)	☐		☐	☐	☐
Molybdenum	☐		☐	☐	☐
Nickel	☐		☐	☐	☐
Selenium	☐		☐	☐	☐
Silver	☐		☐	☐	☐
Thallium	☐		☐	☐	☐
Vanadium	☐		☐	☐	☐
Zinc	☐		☐	☐	☐
Demands					
5-day BOD	☐		☐	☐	☐
5-day Carbonaceous BOD	☒		☒	☐	☐
COD	☐		☐	☐	☐
TOC	☐		☐	☐	☐
Minerals					
Alkalinity, total (CaCO ₃)	☐		☐	☐	☐
Chloride	☐		☐	☐	☐
Fluoride	☐		☐	☐	☐
Hardness, total (CaCO ₃)	☐		☐	☐	☐
Specific conductance (25°C)	☐		☐	☐	☐
Sulfate	☐		☐	☐	☐
Total Dissolved Solids (180°C)	☐		☐	☐	☐
Nutrients					
Ammonia as N	☒		☒	☐	☐
Nitrate as N	☐		☐	☐	☐
Nitrite as N	☐		☐	☐	☐
Orthophosphate as P	☐		☐	☐	☐
Total Kjeldahl-Nitrogen as N	☐		☐	☐	☐
Total Phosphorus as P	☒		☒	☐	☐
Misc. Analytes					
Non-Filterable Residue (TSS)	☒		☒	☐	☐
Oil and Grease	☐		☐	☐	☐
pH	☒		☒	☐	☐
Total Cyanide	☐		☐	☐	☐
Total Phenolics (4-AAP)	☐		☐	☐	☐
Total Residual Chlorine	☐		☐	☐	☐
Total Residual Chlorine (Low Level)	☐		☐	☐	☐
Settleable Solids	☐		☐	☐	☐
Turbidity	☐		☐	☐	☐

Name Cindy A. TomaszewskiSignature CATomaszewskiDate 08/31/2018

Chemistry/Microbiology Analyte Checklist

DMR-QA Study 38

Analyte Test	Test Required	Method Number Used (optional)	Laboratory's Graded Result		Analyte determined by state-certified lab*
			Acceptable	Not Acceptable (Corrective Action Required)	
Microbiology					
<i>E. coli</i> , MF or MPN	☐		☐	☐	☐
Fecal Coliform, MF or MPN	☒		☒	☐	☐
Total Coliform, MF or MPN	☐		☐	☐	☐
Trace Metals					
Aluminum	☐		☐	☐	☐
Antimony	☐		☐	☐	☐
Arsenic	☐		☐	☐	☐
Barium	☐		☐	☐	☐
Beryllium	☐		☐	☐	☐
Cadmium	☐		☐	☐	☐
Chromium, total	☐		☐	☐	☐
Chromium, hexavalent	☐		☐	☐	☐
Cobalt	☐		☐	☐	☐
Copper	☒		☒	☐	☐
Iron	☐		☐	☐	☐
Lead	☐		☐	☐	☐
Manganese	☐		☐	☐	☐
Mercury	☐		☐	☐	☐
Mercury (Low Level)	☐		☐	☐	☐
Molybdenum	☐		☐	☐	☐
Nickel	☐		☐	☐	☐
Selenium	☐		☐	☐	☐
Silver	☐		☐	☐	☐
Thallium	☐		☐	☐	☐
Vanadium	☐		☐	☐	☐
Zinc	☐		☐	☐	☐
Demands					
5-day BOD	☐		☐	☐	☐
5-day Carbonaceous BOD	☐		☐	☐	☐
COD	☐		☐	☐	☐
TOC	☐		☐	☐	☐
Minerals					
Alkalinity, total (CaCO ₃)	☐		☐	☐	☐
Chloride	☐		☐	☐	☐
Fluoride	☐		☐	☐	☐
Hardness, total (CaCO ₃)	☐		☐	☐	☐
Specific conductance (25°C)	☐		☐	☐	☐
Sulfate	☐		☐	☐	☐
Total Dissolved Solids (180°C)	☐		☐	☐	☐
Nutrients					
Ammonia as N	☐		☐	☐	☐
Nitrate as N	☐		☐	☐	☐
Nitrite as N	☐		☐	☐	☐
Orthophosphate as P	☐		☐	☐	☐
Total Kjeldahl-Nitrogen as N	☐		☐	☐	☐
Total Phosphorus as P	☐		☐	☐	☐
Misc. Analytes					
Non-Filterable Residue (TSS)	☐		☐	☐	☐
Oil and Grease	☐		☐	☐	☐
pH	☐		☐	☐	☐
Total Cyanide	☐		☐	☐	☐
Total Phenolics (4-AAP)	☐		☐	☐	☐
Total Residual Chlorine	☐		☐	☐	☐
Total Residual Chlorine (Low Level)	☐		☐	☐	☐
Settleable Solids	☐		☐	☐	☐
Turbidity	☐		☐	☐	☐

Name Cindy A. TomaszewskiSignature C. A. TomaszewskiDate 08/31/2018

WET Analyte Checklist

DMR-QA Study 38

Analyte Number	Organism / Conditions	Endpoint	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab*
				Acceptable	Not Acceptable (Corrective Action Required)	
Test Code 13 (refer to EPA Method 2000.0)						
754	Fathead minnow (<i>Pimephales promelas</i>) - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 14 (refer to EPA Method 2000.0)						
755	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	LC50	☐	☐	☐	☐
Test Code 15 (refer to EPA Method 1000.0)						
756	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	NOEC SURVIVAL	☐	☐	☐	☐
808	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	IC25** (ON) GROWTH	☐	☐	☐	☐
810	Fathead minnow (<i>Pimephales promelas</i>) - MHSF	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 16 (refer to EPA Method 1000.0)						
759	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	NOEC SURVIVAL	☐	☐	☐	☐
812	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	IC25** (ON) GROWTH	☐	☐	☐	☐
814	Fathead minnow (<i>Pimephales promelas</i>) - 20% DMW	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 19 (refer to EPA Method 2002.0)						
764	<i>Ceriodaphnia dubia</i> - MHSF 25°C	LC50	☒	☒	☐	☐
Test Code 20 (refer to EPA Method 2002.0)						
765	<i>Ceriodaphnia dubia</i> - 20% DMW 25°C	LC50	☐	☐	☐	☐
Test Code 21 (refer to EPA Method 1002.0)						
766	<i>Ceriodaphnia dubia</i> - MHSF	NOEC SURVIVAL	☒	☒	☐	☐
767	<i>Ceriodaphnia dubia</i> - MHSF	IC25** REPRODUCTION	☒	☒	☐	☐
768	<i>Ceriodaphnia dubia</i> - MHSF	NOEC REPRODUCTION	☒	☒	☐	☐
Test Code 22 (refer to EPA Method 1002.0)						
769	<i>Ceriodaphnia dubia</i> - 20% DMW	NOEC SURVIVAL	☐	☐	☐	☐
770	<i>Ceriodaphnia dubia</i> - 20% DMW	IC25** REPRODUCTION	☐	☐	☐	☐
771	<i>Ceriodaphnia dubia</i> - 20% DMW	NOEC REPRODUCTION	☐	☐	☐	☐
Test Code 32 (refer to EPA Method 2021.0)						
788	<i>Daphnia magna</i> - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 38 (refer to EPA Method 2021.0)						
794	<i>Daphnia pulex</i> - MHSF 25°C	LC50	☐	☐	☐	☐
Test Code 42 (refer to EPA Method 2007.0)						
798	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 43 (refer to EPA Method 1007.0)						
799	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>)	NOEC SURVIVAL	☐	☐	☐	☐
816	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>)	IC25** (ON) GROWTH	☐	☐	☐	☐
818	Mysid (<i>Americamysis bahia</i> , <i>Mysidopsis bahia</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 44 (refer to EPA Method 2006.0)						
803	Inland silverside (<i>Menidia berylina</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 45 (refer to EPA Method 1006.0)						
824	Inland silverside (<i>Menidia berylina</i>)	NOEC SURVIVAL	☐	☐	☐	☐
825	Inland silverside (<i>Menidia berylina</i>)	IC25** (ON) GROWTH	☐	☐	☐	☐
826	Inland silverside (<i>Menidia berylina</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐
Test Code 46 (refer to EPA Method 2004.0)						
804	Sheepshead minnow (<i>Cyprinodon variegatus</i>) 25°C	LC50	☐	☐	☐	☐
Test Code 47 (refer to EPA Method 1004.0)						
805	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	NOEC SURVIVAL	☐	☐	☐	☐
820	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	IC25** (ON) GROWTH	☐	☐	☐	☐
822	Sheepshead minnow (<i>Cyprinodon variegatus</i>)	NOEC (ON) GROWTH	☐	☐	☐	☐

Name Cindy A. TomaszewskiSignature *CATomaszewski*Date 08/31/2018

*See Footnote on page 2.

**Preferred endpoint for DMR-QA performance test reporting.



A Waters Company

Study: DMR-QA 38
NPDES Permit #: MI0020257
Laboratory Name: Cadillac WWTP

Inorganic Results





A Waters Company

DMR-QA 38 Final Report

NPDES Permit #: M10020257
Permit Holder: Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 08/03/18
Study Dates: 03/23/18 - 07/06/18

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Hardness (cat# 580, lot# Q038-507)											
1960	Total Suspended Solids	mg/L	Acceptable	83.6	86.4	72.7 - 98.1	SM 2540-D-2011 2011	86.9	3.54	M101099	DMRQA38
DMRQA pH (cat# 577, lot# Q038-977)											
1900	pH	S.U.	Acceptable	8.11	8.16	7.96 - 8.36	SM 4500-H+ B-2011 2011	8.14	0.108	M101099	DMRQA38
DMRQA Simple Nutrients (cat# 584, lot# Q038-505)											
1515	Ammonia as N	mg/L	Acceptable	6.38	5.36	4.16 - 6.59	SM 4500-NH3 D-2011 2011	5.38	0.435	M101099	DMRQA38
DMRQA Complex Nutrients (cat# 579, lot# Q038-525)											
1910	Total phosphorus as P	mg/L	Acceptable	7.97	8.66	7.22 - 10.0	SM 4500-P E-2011 2011	8.67	0.445	M101099	DMRQA38
DMRQA Demand (cat# 578, lot# Q038-516)											
1555	CBOD	mg/L	Acceptable	116	96.7	45.0 - 148	SM 5210 B-2011 2011	100	20.6	M101099	DMRQA38



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





A Waters Company

Study: DMR-QA 38
NPDES Permit #: MI0020257
Laboratory Name: Cadillac WWTP

Microbiology Results





A Waters Company

DMR-QA 38 Final Report

NPDES Permit #: MI0020257
Permit Holder: Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 08/03/18
Study Dates: 03/23/18 - 07/06/18

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
2530	Fecal Coliforms (MPN)	MPN/100mL	Acceptable	1732 g	2300	634 - 2960	SM 9223 g (Coliform Quant. Tray)-2004 2004	1370	388	MI00082	DMRQA38

DMRQA WasteWater™ Coliform Microbe™ (cat# 576, lot# Q038-083)

Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





A Waters Company

Study: **DMR-QA 38**
NPDES Permit #: **MI0020257**
Laboratory Name: **Cadillac WWTP**

WET Results





DMR-QA 38 Final Report

A Waters Company

NPDES Permit #: MI0020257

Permit Holder: Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 08/03/18
Study Dates: 03/23/18 - 07/06/18

Analyte Code	Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
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DMRWet Fathead minnow (Test Code 15) (cat# WET004, lot# Q038-004)
7-day Chronic, Daily Renewal, MHSF
Potassium chloride

0808	IC25 (ON) Growth	Acceptable	56.2	42.8	13.4 - 72.3	EPA 1000.0 Chronic 2002	42.8	14.7	MI01183	DMRWet38
0810	NOEC (ON) Growth	Acceptable	50	25.0	12.5 - 50.0	EPA 1000.0 Chronic 2002			MI01183	DMRWet38
0756	NOEC Survival	Acceptable	50	25.0	12.5 - 50.0	EPA 1000.0 Chronic 2002			MI01183	DMRWet38

DMRWet Ceriodaphnia dubia (Test Code 21) (cat# WET010, lot# Q038-010)
3-Brood Chronic, Daily Renewal, MHSF
Potassium chloride

0767	IC25 Reproduction	Acceptable	30.7	37.2	19.6 - 54.8	EPA 1002.0 Chronic 2002	37.2	8.80	MI01183	DMRWet38
0768	NOEC Reproduction	Acceptable	25	25.0	12.5 - 50.0	EPA 1002.0 Chronic 2002			MI01183	DMRWet38
0766	NOEC Survival	Acceptable	50	50.0	25.0 - 100	EPA 1002.0 Chronic 2002			MI01183	DMRWet38



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com



6390 Joyce Drive
 # 100
 Golden, CO 80403
 Phone 303-940-0033
 Fax 866-283-0269
 www.phenova.com

DMRQA38 Graded Results Report

Study: WP0418 for DMRQA38

Opening Date: April 2, 2018 - Closing Date: May 17, 2018

EPA Lab ID: MI00005

Laboratory: Pace Analytical - Grand Rapids
 5560 Corporate Exchange Ct SE
 Grand Rapids, MI 49512
 UNITED STATES
 Ms. Lucy Asbury, Quality Manager
 616-975-4509

NPDES Permit ID: MI0020257

Permittee Name: CADILLAC WWTP

Mercury Low Level (PT-HGLL-WP)				Lot #: 8188-47		
NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result
1095	Mercury	10237204	EPA 1631E	ng/L	58.6	51.0
					42.9 - 73.6	Acceptable



phenovaTM
Certified Reference Materials

A Phenomenex[®]
Company



6390 Joyce Drive
100
Golden, CO 80403

Phone 303-940-0033
Fax 866-283-0269
www.phenova.com

NPDES Number: MI0020257

Veldkamp, Christine (DEQ)

From: Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Sent: Friday, August 31, 2018 4:58 PM
To: Veldkamp, Christine (DEQ)
Subject: DMR-QA 38 (MI0020257)
Attachments: DMR-QA_Study38_2018_MI0020257.CADWTPpg12signed.pdf; DMR-QA_Study38_2018_MI0020257.CADWTPpg12signed.pdf; DMR-QA_Study38_2018_MI0020257.ERMpg13signed.pdf; DMR-QA_Study38_2018_MI0020257.PACEpg12signed.pdf; DMR-QA_Study38_2018_MI0020257pg10.11signed.pdf

Christine –

Attached is form EPA 6400-01 and our signed checklists. You should have rec'd graded results directly from the PT providers on behalf of all four labs we use.

If you have any questions, just send a reply or give me a call. Thank you,

Cindy A. Tomaszewski

Lab Supervisor

City of Cadillac – Utilities Dept.

231-775-2368 (lab)

231-878-6230 (cell)